

Objective

To design a CMOS inverter using Cadence Virtuoso, create its schematic, symbol, and layout, verify the design using DRC and LVS, and analyze its switching characteristics through transient simulation.

Theory of CMOS Inverter

CMOS inverter consists of one PMOS and one NMOS transistor.

PMOS is connected to VDD.

NMOS is connected to GND.

Both gates are connected together as input.

Both drains are connected together as output.

It produces the complement of the input.

CMOS has very low static power consumption because only one transistor conducts in steady state.

Design specifications

Technology: GPDK180 (or your technology)

Supply Voltage (VDD): 1.8 V

NMOS Width: 1 μm

PMOS Width: 2 μm

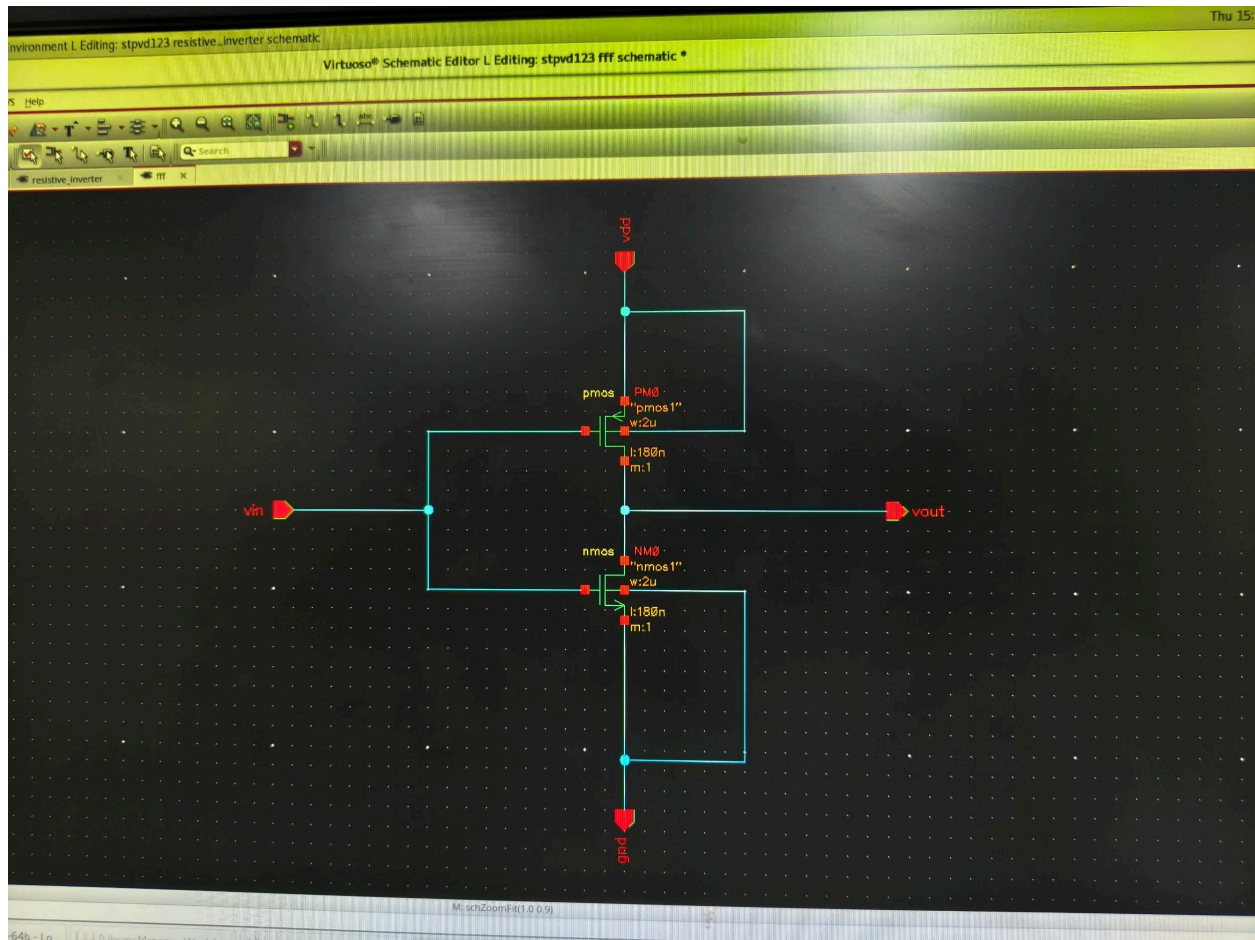
Channel Length: 180 nm

Simulation Type: Transient

Input Signal: Pulse

Schematic

The CMOS inverter schematic was created using one PMOS and one NMOS transistor. Both gates are connected to the input, while the drains are connected together to form the output.



Pre layout simulation waveform

Obtained from the schematic before layout. It assumes ideal interconnections and shows the basic functionality of the CMOS inverter.

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Virtuoso® Analog Design Environment L Editing: stpvd123 cmosinv_test schematic

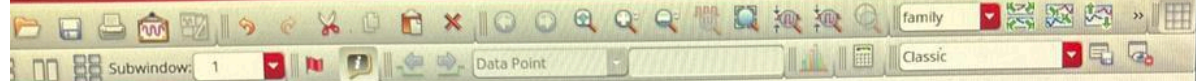
Window PVS Help



Virtuoso (R) Visualization & Analysis XL

Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence



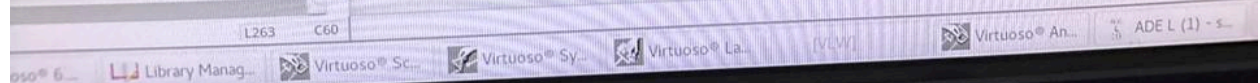
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response Thu Jul 2 18:07:11



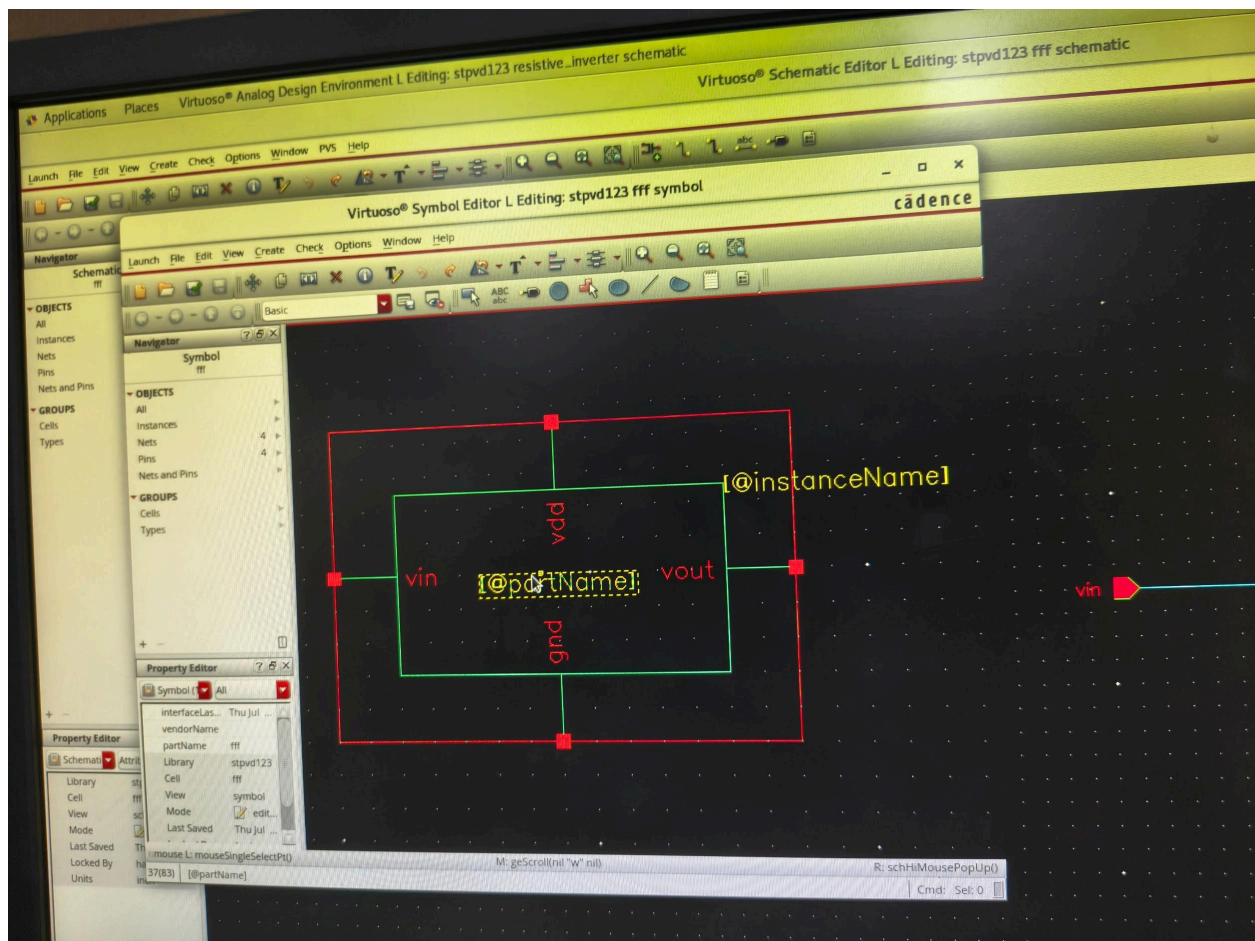
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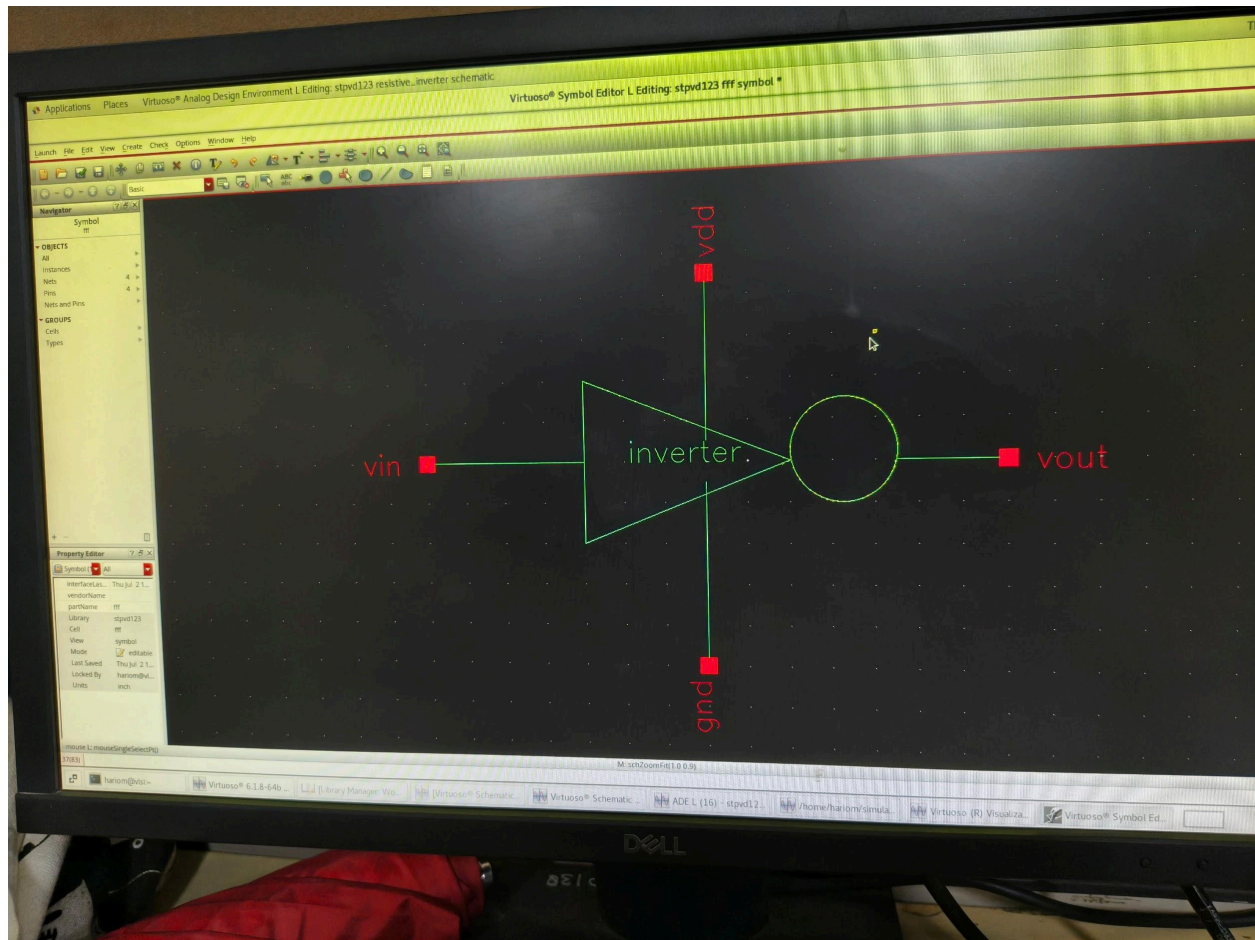
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Symbol

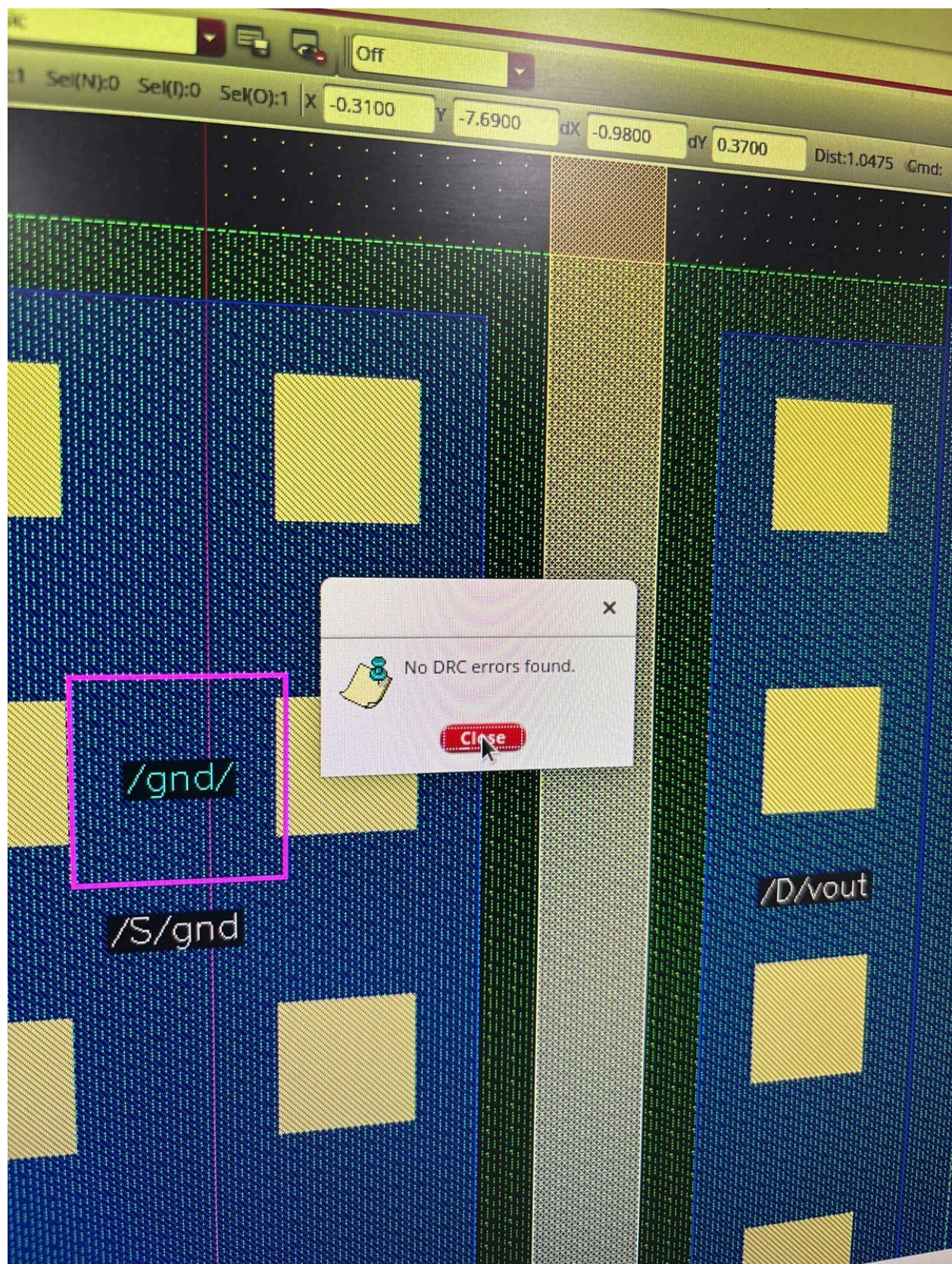
A custom symbol was generated from the schematic. This symbol can be reused in larger circuit designs.





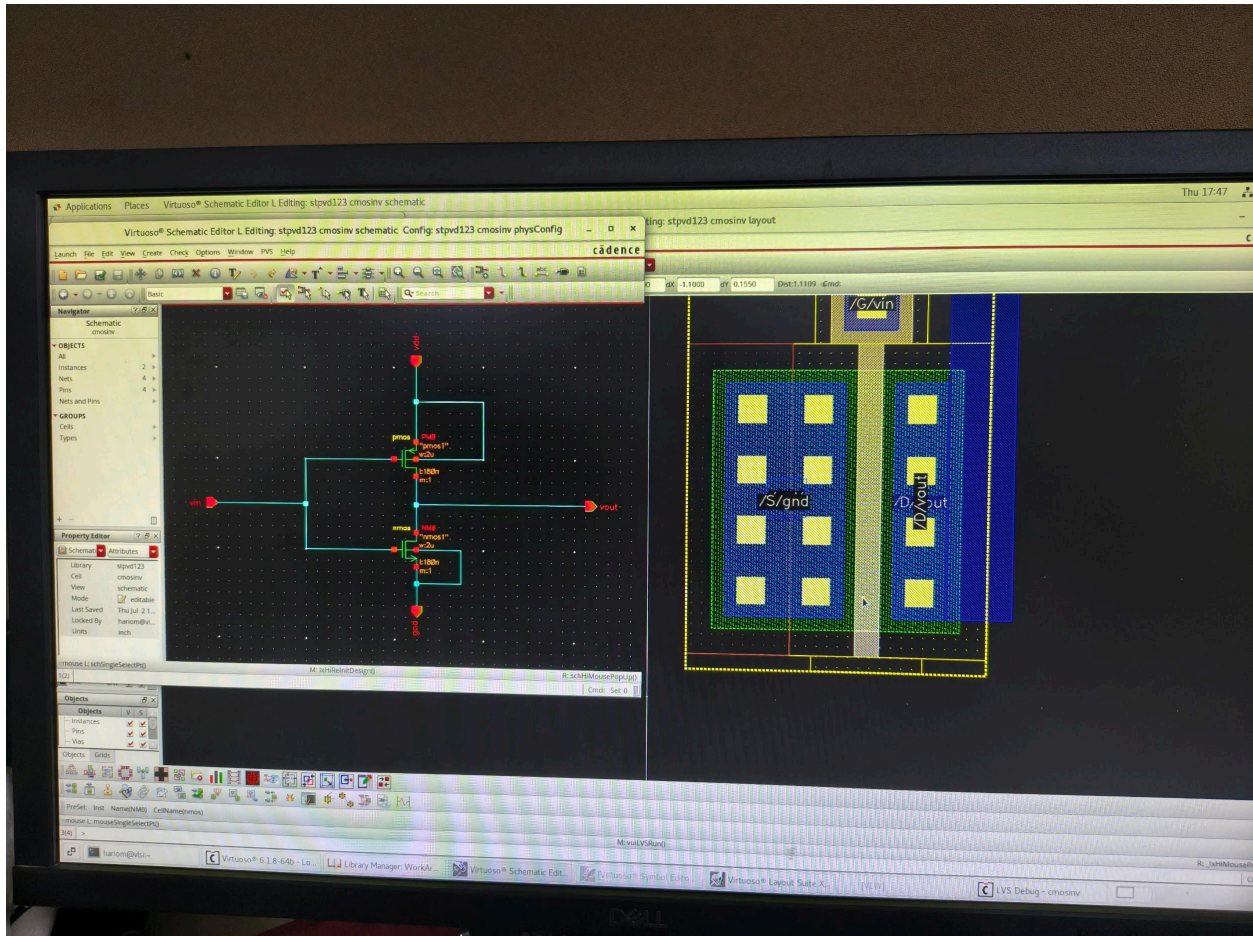
Layout

The physical layout was designed following CMOS layout rules. Metal, diffusion, polysilicon, and contacts were placed according to the technology design rules.



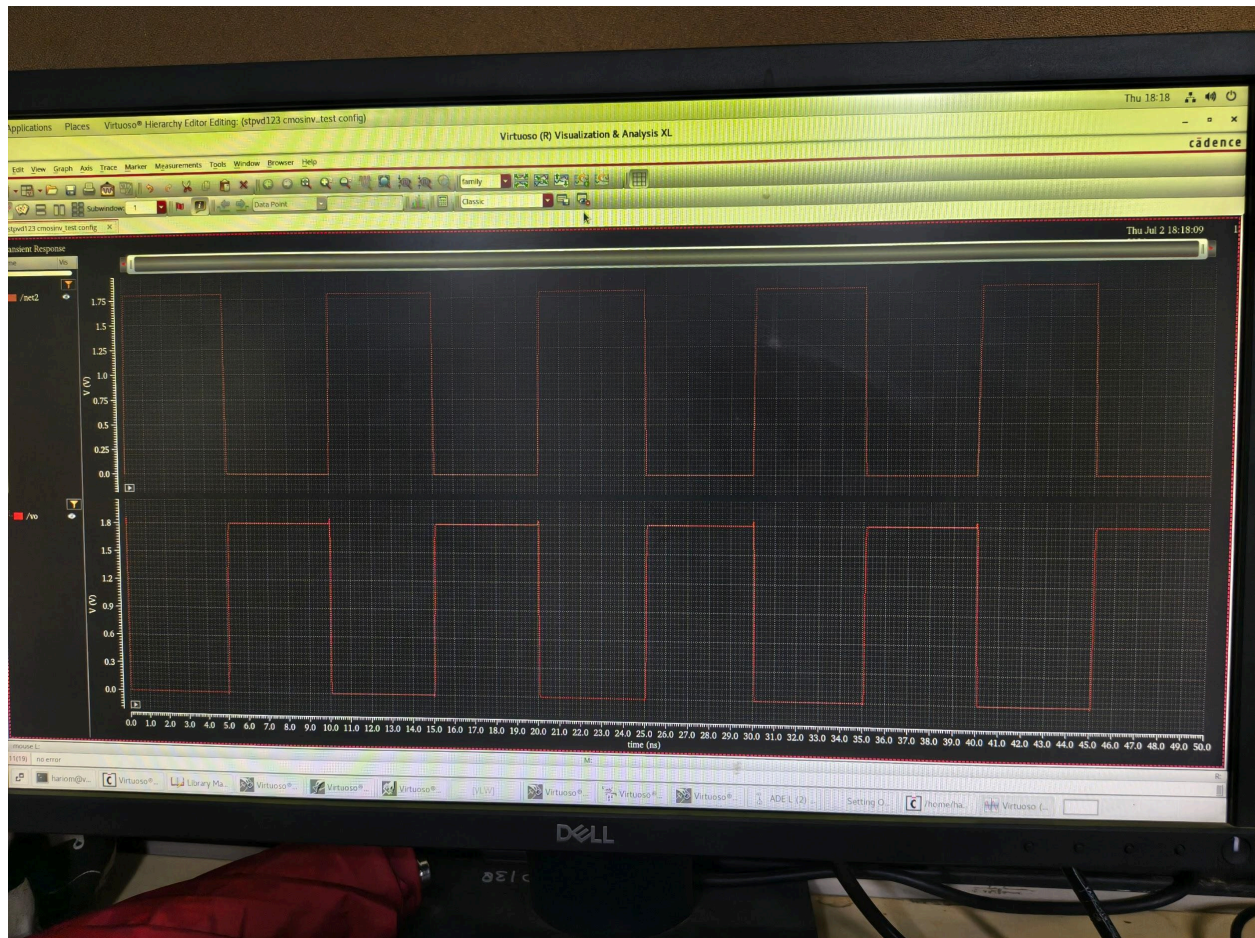
LVS result

Layout Versus Schematic (LVS) confirms that the layout is electrically equivalent to the schematic. The LVS result showed that both matched successfully.



Post layout simulation waveform

Obtained after layout extraction (after DRC/LVS). It includes parasitic resistances and capacitances from the layout, so you may observe a small propagation delay or slightly slower rise/fall times.



Observations

CMOS inverter performs NOT operation.
Output is always the inverse of input.
Static power consumption is very low.
DRC completed with zero errors.
LVS confirmed the layout matches the schematic.

Conclusion

The CMOS inverter was successfully designed and verified using Cadence Virtuoso. The schematic, symbol, and layout were completed successfully. DRC and LVS were passed without errors, and transient simulation verified the correct inverter operation. The design meets the required specifications and is ready for use in larger CMOS circuits.